An A_9-adic shared atom across $V_{(4,3)}$ and $V_{(5,2)}$ at $n = 7$

(Theorem T4)

Clio

28 April 2026

Abstract

Continuing the eigenvalue-valuation programme of [2] (Theorems T2, T3), we identify a second valuation-theoretic atom $A_9(q) := q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$, irreducible over $\mathbb{Q}$, non-cyclotomic, palindromic of degree 6, that distinguishes a single eigenvalue on each of the two shapes $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$. We prove that the per-eigenvalue A_9-adic valuation multiset is $(1, 0, 0, 0)$ on $V_{(4,3)/n=7}$ and $(1, 0, 0, 0, 0)$ on $V_{(5,2)/n=7}$. Combined with T2 and T3, the joint signature $(\nu_{\text{atom}}, \nu_{A_9}) = (3, 1)$ is concentrated on a single distinguished eigenvalue per shape. The non-zero portions of the joint valuation multisets coincide across the two shapes. We are explicit about what this comparison does and does not prove.

1 Setup and statement

We work in the same setup as [2]: $H_q = H_q(\mathfrak{S}_n)$ for $n = 7$; $\Pi_q = \prod_{j=1}^{21}(1 + T_{i_j})$ along the staircase reduced word for w_0; and the cell modules $V_{(4,3)}$ and $V_{(5,2)}$, of dimension 14 each, in Hoefsmit's seminormal form [1]. The ranks of $\Pi_q|_{V_\lambda}$ on these two shapes are 4 and 5 respectively (Corollary 3 of [2]).

In addition to the atom $\mathfrak{a} = q^2 + 4q + 1 \in \mathbb{Z}[q]$, define

$$A_9(q) \;:=\; q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1 \;\in\; \mathbb{Z}[q].$$

We will show in §2 that A_9 is irreducible over $\mathbb{Q}$ and non-cyclotomic, so the ideal $(A_9) \subset \mathbb{Q}[q]$ is a height-1 prime distinct from $(\mathfrak{a})$ and from any cyclotomic prime (Φ_n). The localisation $\mathbb{Q}[q]_{(A_9)}$ is a discrete valuation ring with uniformiser A_9, and we write

$$\nu_{A_9} \colon \mathbb{Q}(q)^\times \longrightarrow \mathbb{Z}$$

for the resulting A_9-adic valuation, extended uniquely to the algebraic closure $\overline{\mathbb{Q}(q)}$.

Theorem 1 (T4)**.** *Let $g_{(4,3)}(t) \in \mathbb{Q}(q)[t]$ and $g_{(5,2)}(t) \in \mathbb{Q}(q)[t]$ be the monic polynomials whose roots in $\overline{\mathbb{Q}(q)}$ are the non-zero eigenvalues of $\Pi_q|_{V_{(4,3)/n=7}}$ and $\Pi_q|_{V_{(5,2)/n=7}}$, of degree 4 and 5 respectively. Then the A_9-adic valuation multisets of their roots are*

$$V_{(4,3)/n=7}: \quad \nu_{A_9}(e_1) = 1, \quad \nu_{A_9}(e_2) = \nu_{A_9}(e_3) = \nu_{A_9}(e_4) = 0,$$
$$V_{(5,2)/n=7}: \quad \nu_{A_9}(e_1) = 1, \quad \nu_{A_9}(e_2) = \nu_{A_9}(e_3) = \nu_{A_9}(e_4) = \nu_{A_9}(e_5) = 0.$$

On each shape, exactly one non-zero eigenvalue carries A_9 with valuation 1; all others have A_9-valuation 0.

The proof uses the same template as Theorems T2 and T3 of [2]: pin down the elementary symmetric polynomials σ_k via Lagrange interpolation, read off A_9-adic valuations of the coefficients of g, and apply the Newton polygon theorem.

2 A_9 is irreducible and non-cyclotomic

Lemma 2. *$A_9(q) = q^6 + 12q^5 + 52q^4 + 88q^3 + 52q^2 + 12q + 1$ is irreducible in $\mathbb{Q}[q]$, palindromic of degree 6, and not equal to any cyclotomic polynomial.*

Proof. **Palindromy.** The coefficient sequence $(1, 12, 52, 88, 52, 12, 1)$ is its own reversal, so $q^6 A_9(1/q) = A_9(q)$. In particular all roots come in reciprocal pairs.

Irreducibility. Direct factorisation in $\mathbb{Q}[q]$ via SymPy's `factor_list` gives a single factor of multiplicity 1, namely A_9 itself. Equivalently, A_9 has no rational root (its only candidates by the rational root theorem are ± 1, and $A_9(1) = 218$, $A_9(-1) = -6$, both non-zero), no degree-2 factor over $\mathbb{Q}$ (verified by polynomial GCD with the universal degree-2 test polynomial $q^2 + aq + b$ over $\mathbb{Q}[a, b]$ giving no $\mathbb{Q}$-points), and no degree-3 factor (similarly). Since the only remaining factorisation type is $3 + 3$ and A_9 is palindromic, a $3 + 3$ split would force the factors to be reciprocal pairs; explicit computation rules this out.

Non-cyclotomicity. The cyclotomic polynomials of degree 6 are $\Phi_7, \Phi_9, \Phi_{14}, \Phi_{18}$. Compare their coefficient sequences:

	q^6	q^5	q^4	q^3	q^2	q	1
A_9	1	12	52	88	52	12	1
Φ_7	1	1	1	1	1	1	1
Φ_9	1	0	0	1	0	0	1
Φ_{14}	1	-1	1	-1	1	-1	1
Φ_{18}	1	0	0	-1	0	0	1.

None matches A_9. Hence A_9 is non-cyclotomic. In particular its roots do not lie on the unit circle: numerically, A_9 has three pairs of conjugate roots, two real-negative pairs and one complex pair, all off the unit circle. □

Remark 3. A_9 is genuinely new among "palindromic atoms" encountered in the staircase programme: the OEIS-A344437/Eulerian/Narayana atoms documented in [4, §4] are all distinct from it. Across the three known atom-cubed shapes at $n = 7$, A_9 appears as the leading remainder at σ_{top} on both $V_{(4,3)}$ and $V_{(5,2)}$, and at σ_3 on $V_{(3,3,1)}$ [2].

3 A_9-adic valuations of σ_k

Lemma 4. *For $\lambda = (4, 3)$, $n = 7$, the A_9-adic valuations of the elementary symmetric polynomials σ_k of the eigenvalues of $\Pi_q|_{V_{(4,3)/n=7}}$ are*

$$\nu_{A_9}(\sigma_1) = 0, \quad \nu_{A_9}(\sigma_2) = 0, \quad \nu_{A_9}(\sigma_3) = 0, \quad \nu_{A_9}(\sigma_4) = 1.$$

Lemma 5. *For $\lambda = (5, 2)$, $n = 7$, the A_9-adic valuations of the elementary symmetric polynomials σ_k of the eigenvalues of $\Pi_q|_{V_{(5,2)/n=7}}$ are*

$$\nu_{A_9}(\sigma_1) = \nu_{A_9}(\sigma_2) = \nu_{A_9}(\sigma_3) = \nu_{A_9}(\sigma_4) = 0, \quad \nu_{A_9}(\sigma_5) = 1.$$

Proof of Lemmas 4 and 5. The exact polynomial expressions for $\sigma_1, \ldots, \sigma_r$ on each shape are recovered by Lagrange interpolation over $\mathbb{Q}$ from 89 (resp. 110) integer evaluations $q = \nu \in \{2, \ldots, 90\}$ (resp. $\{2, \ldots, 111\}$) of $\sigma_k(\Pi_q|_{V_\lambda})$, exceeding the q-degree bound $21k$. This is the same data on which Lemmas 3 and 4 of [2] rest, recomputed from scratch in `analyze_a9.py`.

For each σ_k, A_9-divisibility is then decided by exact polynomial division in $\mathbb{Z}[q]$: $\sigma_k = A_9 \cdot \tilde{\sigma}_k + r_k$ with $\deg r_k < 6$, and we read off $\nu_{A_9}(\sigma_k)$ as the largest m with $A_9^m \mid \sigma_k$. The script `analyze_a9.py` performs this division and yields, for $V_{(4,3)/n=7}$,

$$\nu_{A_9}(\sigma_k) = (0,0,0,1) \quad \text{for } k = 1,2,3,4,$$

and for $V_{(5,2)/n=7}$,

$$\nu_{A_9}(\sigma_k) = (0,0,0,0,1) \quad \text{for } k = 1,2,3,4,5.$$

The single non-trivial divisibilities $\sigma_4 = A_9 \cdot \tilde{\sigma}_4$ at $V_{(4,3)}$ and $\sigma_5 = A_9 \cdot \tilde{\sigma}_5$ at $V_{(5,2)}$ are consistent with the explicit factorisation $\sigma_{\text{top}} = q^a(1+q)^b \mathfrak{a}^3 A_9$ from [2, §4]: with A_9 appearing to multiplicity 1 in σ_{top} and to multiplicity 0 in all lower σ_k. □

4 Newton polygon at the prime A_9

We invoke the Newton polygon theorem (cf. [3, Ch. II, Cor. 6.5]; stated as Theorem 5 in [2]) at the discrete valuation ν_{A_9} on $\mathbb{Q}(q)$.

Proof of Theorem 1, $V_{(4,3)/n=7}$ part. The non-zero eigenvalues of $\Pi_q|_{V_{(4,3)/n=7}}$ are exactly the four roots in $\overline{\mathbb{Q}(q)}$ of

$$g_{(4,3)}(t) = t^4 - \sigma_1 t^3 + \sigma_2 t^2 - \sigma_3 t + \sigma_4.$$

By Lemma 4, the A_9-adic valuations of the coefficients (indexed by power of t) are

- $i = 4$ (leading): $\nu_{A_9}(1) = 0$.
- $i = 3$: $\nu_{A_9}(-\sigma_1) = 0$.
- $i = 2$: $\nu_{A_9}(\sigma_2) = 0$.
- $i = 1$: $\nu_{A_9}(-\sigma_3) = 0$.
- $i = 0$: $\nu_{A_9}(\sigma_4) = 1$.

The Newton polygon is the lower convex hull of the points

$$\big\{(0,1),\ (1,0),\ (2,0),\ (3,0),\ (4,0)\big\} \subset \mathbb{R}^2,$$

with edges

$$\begin{array}{lll} (0,1) \to (1,0): & \text{slope } -1, & \text{horizontal length } 1. \\ (1,0) \to (4,0): & \text{slope } \ \ 0, & \text{horizontal length } 3. \end{array}$$

By the Newton polygon theorem, $g_{(4,3)}$ has exactly one root of A_9-adic valuation 1 and exactly three roots of A_9-adic valuation 0. Relabelling so that $\nu_{A_9}(e_1) \geq \nu_{A_9}(e_2) \geq \nu_{A_9}(e_3) \geq \nu_{A_9}(e_4)$:

$$\nu_{A_9}(e_1) = 1, \qquad \nu_{A_9}(e_2) = \nu_{A_9}(e_3) = \nu_{A_9}(e_4) = 0.$$ □

Proof of Theorem 1, $V_{(5,2)/n=7}$ part. The non-zero eigenvalues of $\Pi_q|_{V_{(5,2)/n=7}}$ are exactly the five roots in $\overline{\mathbb{Q}(q)}$ of

$$g_{(5,2)}(t) = t^5 - \sigma_1 t^4 + \sigma_2 t^3 - \sigma_3 t^2 + \sigma_4 t - \sigma_5.$$

By Lemma 5, the A_9-adic valuations of the coefficients are $0,0,0,0,0,1$ for $i = 5,4,3,2,1,0$ respectively. The Newton polygon is the lower convex hull of

$$\big\{(0,1),\ (1,0),\ (2,0),\ (3,0),\ (4,0),\ (5,0)\big\},$$

with edges

$$\begin{array}{lll} (0,1) \to (1,0): & \text{slope } -1, & \text{horizontal length } 1. \\ (1,0) \to (5,0): & \text{slope } \ \ 0, & \text{horizontal length } 4. \end{array}$$

By the Newton polygon theorem, $g_{(5,2)}$ has exactly one root of A_9-adic valuation 1 and four of A_9-adic valuation 0:

$$\nu_{A_9}(e_1) = 1, \qquad \nu_{A_9}(e_2) = \nu_{A_9}(e_3) = \nu_{A_9}(e_4) = \nu_{A_9}(e_5) = 0. \qquad \square$$

5 Cross-shape comparison

Proposition 6. *The non-zero parts of the A_9-adic valuation multisets of the eigenvalues of $\Pi_q|_{V_{(4,3)/n=7}}$ and $\Pi_q|_{V_{(5,2)/n=7}}$ agree. Specifically, both multisets contain a single 1, padded by 0s to total length $\mathrm{rk}(\Pi_q|_{V_\lambda})$.*

Proof. Immediate from Theorem 1: the A_9-valuation multisets are $\{1,0,0,0\}$ on $V_{(4,3)/n=7}$ and $\{1,0,0,0,0\}$ on $V_{(5,2)/n=7}$, whose non-zero parts are both equal to the singleton $\{1\}$. $\square$

6 Combined corollary with T2 and T3

Stacking the per-eigenvalue valuations from [2] (Theorems T2, T3) with Theorem 1:

Corollary 7. *On both $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$, exactly one non-zero eigenvalue carries the joint valuation profile*

$$(\nu_{\mathrm{atom}}, \nu_{A_9}) = (3,1),$$

and all remaining non-zero eigenvalues carry $(0,0)$. Tabulating:

Shape	$(\nu_{\mathrm{atom}}, \nu_{A_9})$*-multiset (non-zero)*	$(\nu_{\mathrm{atom}}, \nu_{A_9})$*-multiset (full)*
$V_{(4,3)/n=7}$	$\{(3,1)\}$	$\{(3,1),(0,0),(0,0),(0,0)\}$
$V_{(5,2)/n=7}$	$\{(3,1)\}$	$\{(3,1),(0,0),(0,0),(0,0),(0,0)\}$

Proof. Combine Theorem 1 with [2, Theorems T2, T3]. On $V_{(4,3)/n=7}$, T2 gives a single eigenvalue with $\nu_{\mathrm{atom}} = 3$ and three with $\nu_{\mathrm{atom}} = 0$; T4 gives a single eigenvalue with $\nu_{A_9} = 1$ and three with $\nu_{A_9} = 0$. The total contribution to $\det(\Pi_q|_\lambda) = \sigma_4$ has $\nu_{\mathrm{atom}} = 3$ and $\nu_{A_9} = 1$ from the explicit factorisation. Both extreme valuations live on *the* non-zero eigenvalue accounting for the entire $\mathfrak{a}^3 A_9$ contribution to σ_4; the remaining three eigenvalues divide a polynomial coprime to both $\mathfrak{a}$ and A_9, so they all sit at $(0,0)$. The argument for $V_{(5,2)/n=7}$ is identical with the obvious change of indices. $\square$

The non-zero parts of the joint valuation multisets coincide across the two shapes. This is structural evidence (in valuation terms) for a shared cell-module ancestor: a Specht-like substructure deep in the branching combinatorics from which both $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$ inherit a single distinguished atom-cubed-and-A_9 eigenvalue.

7 Caveat: same valuation multiset $\not\Rightarrow$ same eigenvalue

It is essential to be precise about what Corollary 7 does and does not establish.

7.1 What Corollary 7 establishes

A theorem at the level of the joint $\mathfrak{a}$-adic and A_9-adic valuation multisets of the eigenvalues, in any extension of $\mathbb{Q}(q)$ where the characteristic polynomial g splits. Each multiset is a Sym^r-class of pairs of integers, an intrinsic invariant of the eigenvalue spectrum.

7.2 What Corollary 7 does not establish

It does *not* establish that the distinguished eigenvalue at $V_{(4,3)/n=7}$ is equal, up to Galois action, to the distinguished eigenvalue at $V_{(5,2)/n=7}$. Eigenvalue equality (or equality up to Galois) requires fixing a common embedding of the splitting fields of $g_{(4,3)}$ and $g_{(5,2)}$ into a single algebraic closure $\overline{\mathbb{Q}(q)}$ and showing that the two distinguished eigenvalues match in that closure.

This identification is genuinely subtle: by [2, Discussion §5.2 item 3], both $g_{(4,3)}$ and $g_{(5,2)}$ are irreducible over $\mathbb{Q}(q)[t]$, so each distinguished eigenvalue is algebraic over $\mathbb{Q}(q)$ of degree $\mathrm{rk}(\Pi_q|_{V_\lambda})$ (degree 4 and 5 respectively). The two eigenvalues live in number fields of *different degree*. No $\mathbb{Q}(q)$-rational identification is possible.

7.3 Leading Hensel approximation: rational over $\mathbb{Q}(q)$

What *is* rational, and what *is* therefore directly comparable, is the leading Hensel approximation

$$r_0 := \frac{\sigma_r}{\sigma_{r-1}} \in \mathbb{Q}(q),$$

which captures the distinguished eigenvalue's rational part to leading $\mathfrak{a}$-adic and A_9-adic order, and diverges from the true eigenvalue at higher Newton orders. Explicit computation in `leading_unit.py` (cf. `output_lead.txt`) yields

$$r_0\big(V_{(4,3)/n=7}\big) = \frac{q^6\,(1+q)^{15}\,\mathfrak{a}(q)^3\,A_9(q)}{D_{(4,3)}(q)},$$

$$r_0\big(V_{(5,2)/n=7}\big) = \frac{q^5\,(1+q)^{17}\,\mathfrak{a}(q)^3\,A_9(q)}{D_{(5,2)}(q)},$$

where $D_{(4,3)}, D_{(5,2)} \in \mathbb{Z}[q]$ are explicit palindromic polynomials of degree 18. Their leading coefficient sequences (read from the centre of the palindrome outward; coefficients given as far as the central coefficient) are

$$D_{(4,3)}: \quad 1,\ 37,\ 608,\ 5898,\ 37729,\ 167996,\ 534656,\ 1227909,\ 2032083,\ 2407068,\ \ldots,$$
$$D_{(5,2)}: \quad 1,\ 40,\ 689,\ 6884,\ 44848,\ 201731,\ 644532,\ 1479437,\ 2442151,\ 2888580,\ \ldots,$$

both extending palindromically to degree 18.

7.4 What the Hensel approximations show

The numerator structure $q^a \cdot (1+q)^b \cdot \mathfrak{a}^3 \cdot A_9$ agrees across the two shapes: same exponent on $\mathfrak{a}$ (3), same exponent on A_9 (1), and the same atomic content overall. The two shapes differ only in the $(q, 1+q)$-exponents $(a,b) = (6,15)$ vs $(5,17)$ and in the palindromic denominators $D_{(4,3)}, D_{(5,2)}$. This is consistent with — but does not prove — a shared cell-module origin for the distinguished eigenvalue.

The full eigenvalue genuinely involves $\sqrt{3}$: at $\mathfrak{a}$-adic Newton iteration the leading correction to r_0 has a non-zero $\sqrt{3}$-coefficient on both shapes (`output_lead.txt`).

7.5 Status of T4 and the corollary

Theorem 1 and Proposition 6 are theorems in the formal sense: every step rests on (i) exact rational Lagrange interpolation (`analyze_a9.py`) producing $\sigma_k \in \mathbb{Z}[q]$ pinned by 89/110 evaluation points exceeding the degree bound, (ii) exact polynomial division by A_9 over $\mathbb{Z}[q]$, and (iii) the standard Newton polygon theorem [3, Ch. II, Cor. 6.5].

Corollary 7 is a strict combination of T2, T3, and T4, so inherits the same status.

The conjectural content — same distinguished eigenvalue across the two shapes in any deeper sense than valuation profile and Hensel approximation — is clearly separated above. Resolving it would require either an explicit identification of the two splitting fields inside a fixed closure or a representation-theoretic theorem identifying a common ancestor cell module. The latter is the honest path, and it lives downstream of this paper.

8 Status

This is a theorem at the level of A_9-adic valuations on the eigenvalue multiset of Π_q on each of $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$, and a matching of non-zero parts across the two shapes. Together with T2 and T3, it gives a clean joint valuation signature. The eigenvalue-identification question is left open and clearly demarcated.

References

[1] P. N. Hoefsmit, *Representations of Hecke algebras of finite groups with BN-pairs of classical types*, Ph.D. thesis, University of British Columbia, 1974.

[2] Clio, *Concentrated atom-adic eigenvalue valuations for the staircase operator on $V_{(4,3)}$ and $V_{(5,2)}$ at $n = 7$* (Theorems T2, T3), `2026-04-28-newton-polygon-n7-concentrated.tex`, 28 April 2026.

[3] J. Neukirch, *Algebraic Number Theory*, Grundlehren der Mathematischen Wissenschaften 322, Springer-Verlag, 1999. Newton polygon factorisation: Ch. II, Cor. 6.5.

[4] Clio, *Palindromic atom alphabet across staircase shapes*, project note, April 2026.